

LECTURE 2: MATH1061/7861
TOPICS: LOGICAL FORMS
WEDNESDAY, FEB. 25, 2026

Learning objectives:

- (1) Appreciate the difficulties in translating between English and mathematics.
- (2) Understand the logical connectives $\sim$ ('not'), $\wedge$ ('and'), and $\vee$ ('or').
- (3) Be able to construct a truth table for a statement form.

Note: Problem set is online.



Great resource for
the semester

A statement is a sentence that is either true or false, but not both.

Q1: Which of the following is a statement?

~~NO~~

(1) Is it going to rain tomorrow?

NO. It's a question.

(2) He is happy.

No. We don't know who is he.
Happy is ambiguous.

(3) 28 of July 2025 is a Wednesday.

Yes.

(4) $67 > 102$.

Yes.

(5) $x = 5y + 2$.

NO. We don't know
what x & y are.

Q2:

Let p be the statement "It is cold in Sydney."

Let q be the statement "It is snowing in New York."

Let r be the statement "It is sunny in Brisbane."

Translate the following into English sentences:

(1) $\sim p \wedge q$ *It is not cold in Sydney,
and it is snowing in New York.*
But

(2) $\sim (q \vee p) \wedge r$ *and*
*[It is not snowing in NY ~~or~~ it is not
cold in Sydney, and it is sunny in Brisbane.]*

(3) $(p \wedge q) \vee r$
• Either it is the case that it is cold in Sydney and snowing in NY,
or it is the case that it is sunny in Brisbane
or it is both cases.

(4) $\sim (p \vee \sim q)$

- It is not the case that: ~~either~~ it is cold in Sydney
or not snowing in New York.
- It is not cold in Sydney and it is snowing in New York.

Notice that

$$\sim (p \vee \sim q) \equiv (\sim p) \wedge (\sim(\sim q)) \equiv (\sim p) \wedge q$$

Valid answers
include:

Other proposals from class

Q2 (2)

2) It is not snowing in NY ~~or~~ ^{AND}

it is not cold in Sydney.

It is sunny in Brisbane.

2) It is not snowing in NY and it's
not cold in Sydney, and it's sunny in Brisbane.

2) It is not snowing in NY ~~or~~ ^{or}
or cold in Sydney; However, it is sunny in
Brisbane.

$\sim p \vee \sim q$

versus

$\sim (p \vee q)$

the same as

$\sim p \wedge \sim q$

for Q2 (2)

Other valid answersⁿ include:

- It is neither snowing in NY nor cold in Sydney,
but it is sunny in Brisbane
- It is not snowing in NY, and it is not cold in Sydney.
Additionally, it is sunny in Brisbane.